Econometria Avançada — MPE (Insper)

Exercise list: *Poisson inference and Monte Carlo methods for inference*

Instructions. Seven blocks, from Poisson likelihood theory to MCMC. Each block has (i) textbook exercises reproduced verbatim from the book PDF (statements from Portuguese-language books stay in Portuguese), (ii) an original math question (derivations and proofs, ending with a parameter analysis) and (iii) an original conceptual question in exam form. Every question carries a *Programming* part added to the book's text: solve it in Python (NumPy/SciPy/statsmodels) with the seeds given; the scaffold notebook `Listas/lista-poisson-monte-carlo.ipynb` lists the steps. The script `Listas/poisson-monte-carlo-check.py` reproduces and asserts every number printed in the solutions. Multiple choice (3 points): one correct option, no penalty. True/False (2 points): score = $\max\{0, \text{correct} - \text{wrong}/2\}\times$ item value; blank items do not count. Book abbreviations: B = Bolfarine & Sandoval (2010), G = Greenberg (2008, 1st ed.), I = ISLP (2023), MS = Morettin & Singer (2021). The coal-mining series used throughout (British mine disasters per year, 1851–1962, Jarrett 1979) is

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1851-1878:  4 5 4 1 0 4 3 4 0 6 3 3 4 0 2 6 3 3 5 4 5 3 1 4 4 1 5 5
1879-1906:  3 4 2 5 2 2 3 4 2 1 3 2 2 1 1 1 1 3 0 0 1 0 1 1 0 0 3 1
1907-1934:  0 3 2 2 0 1 1 1 0 1 0 1 0 0 0 2 1 0 0 0 1 1 0 2 3 3 1 1
1935-1962:  2 1 1 1 1 2 4 2 0 0 0 1 4 0 0 0 1 0 0 0 0 0 1 0 0 1 0 1
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($n = 112$ years, 191 disasters; 127 in the first 41 years, 1851–1891).

Block 1 — Poisson likelihood: sufficiency, information, Cramér–Rao, UMVU *(B 2.4, 2.6, 2.5; original)*

Question 1. B 2.4 — $X_1 + 2X_2$ is not sufficient for a Poisson mean.

Source: Bolfarine, H. and Sandoval, M. C. (2010), *Introdução à Inferência Estatística*, Chapter 2, §2.6 Exercícios, Exercício 2.4, p. 32 [PDF p. 38]; the book gives no answer — reproduced verbatim.

2.4. Sejam X_1, X_2 uma amostra aleatória da variável aleatória $X \sim Poisson(\theta)$. Mostre que $T = X_1 + 2X_2$ não é suficiente para θ.

Programming (added, not in the book). Write `cond_pmf(t, theta, w2)` that enumerates every pair (x_1, x_2) with $x_1 + w_2 x_2 = t$ and returns $P(X_1 = x_1, X_2 = x_2 \mid X_1 + w_2 X_2 = t)$. Print it for $T = X_1 + 2X_2$ ($w_2 = 2$) at $t = 2$ and $t = 4$, for $\theta = 1$ and $\theta = 2$. Repeat for $S = X_1 + X_2$ ($w_2 = 1$) at $s = 4$ and check that the result is the Binomial$(4, 1/2)$ p.m.f. for both values of θ.

Solution:

Question 2. B 2.6 — the information identity.

Source: Bolfarine, H. and Sandoval, M. C. (2010), *Introdução à Inferência Estatística*, Chapter 2, §2.6 Exercícios, Exercício 2.6, p. 33 [PDF p. 39]; the book gives no answer — reproduced verbatim.

2.6. Seja $f(x|\theta)$ uma função densidade para a qual as condições de regularidade estão satisfeitas. Mostre que

$$E\left[\left(\frac{\partial \log f(X|\theta)}{\partial \theta}\right)^2\right] = -E\left[\frac{\partial^2 \log f(X|\theta)}{\partial \theta^2}\right].$$

Programming (added, not in the book). (i) For $X \sim$ Poisson(θ) with $\theta = 3$, compute $E[U]$, $E[U^2]$ and $-E[\partial^2 \log f/\partial\theta^2]$, with $U = \partial \log f(X|\theta)/\partial\theta$, by summing over $x = 0, \ldots, 399$, and compare with $1/\theta$. (ii) Do the same for $X \sim U(0, \theta)$ with $\theta = 2$, by numerical integration (`scipy.integrate.quad`), and explain the result.

Solution:

Question 3. B 2.5 — the Cramér–Rao bound for $g(\theta)$.
Source: Bolfarine, H. and Sandoval, M. C. (2010), *Introdução à Inferência Estatística*, Chapter 2, §2.6 Exercícios, Exercício 2.5, p. 33 [PDF p. 39]; the book gives no answer — reproduced verbatim.

2.5. Sejam $X_1, \ldots, X_n$ uma amostra aleatória da variável aleatória X com função de densidade (ou de probabilidade) $f(x|\theta)$ para a qual as condições de regularidade estão satisfeitas. Seja $\hat{\gamma}$ um estimador não viciado para $g(\theta)$. Mostre que

$$Var(\hat{\gamma}) \geq \frac{(g'(\theta))^2}{nE\left[\left(\frac{\partial \log f(X|\theta)}{\partial \theta}\right)^2\right]}.$$

Programming (added, not in the book). Take $X \sim \text{Poisson}(\theta)$, $g(\theta) = \theta$, $\theta = 2$, $n = 10$. With `np.random.default_rng(3)` draw 200,000 samples of size n; for each compute $\overline{X}$ and $S^2 = \sum_i (X_i - \overline{X})^2/(n-1)$ (both unbiased for θ). Print the simulated means and variances of both estimators next to the bound θ/n and to the exact $\text{Var}(S^2) = \theta/n + 2\theta^2/(n-1)$.

Solution:

Question 4. Original math question — estimating the probability of an accident-free day.
Source: Original question (not from a book).

The Federal Highway Police (PRF) counts serious accidents per day on a stretch of the BR-116. Model the daily counts $X_1, \ldots, X_n$ as i.i.d. Poisson(θ), $\theta > 0$, and let $T = \sum_{i=1}^n X_i$. The indicator the PRF reports is $\psi = P(X = 0) = e^{-\theta}$.

(a) Find the maximum likelihood estimator $\hat{\psi}$ of ψ, its exact expectation, and the sign of its bias for every $n \geq 1$ and $\theta > 0$.

(b) Find the uniformly minimum-variance unbiased estimator (UMVU) $\tilde{\psi}$ of ψ. Prove every step (sufficiency and completeness of T may be quoted from B, Exemplo 2.5.4).

(c) Find $\text{Var}(\tilde{\psi})$ in closed form and compare it with the Cramér–Rao bound for $g(\theta) = e^{-\theta}$. Is there an efficient unbiased estimator of ψ for finite n? Prove your answer.

(d) Parameter analysis: let $\rho(\theta, n) = \text{Var}(\tilde{\psi})/\text{CR bound}$. Prove the signs of $\partial\rho/\partial\theta$ and of the change in ρ with n, find the limits $n \to \infty$, $\theta \to 0$ and the value at $n = 1$, and interpret them for the PRF.

Programming (added, not in the book). With $\theta = 1.5$, $n = 10$ and `np.random.default_rng(4)`, simulate 200,000 samples; print the simulated mean of $\hat{\psi}$ against (a), the mean and variance of $\tilde{\psi}$ against (b)–(c), the bound, and the mean squared error of $\hat{\psi}$. Then print $\rho(1.5, n)$ for $n = 1, 2, 5, 10, 50, 1000$.

Solution:

Question 5. Original conceptual question — sufficiency, information and optimal estimators (10 points).

Source: Original question (not from a book).

The PRF models the daily number of accidents on a stretch of road as $X_1, \ldots, X_n$ i.i.d. Poisson(θ), $T = \sum_i X_i$.

a) (3 points) Which statement about T is correct?

A. T is sufficient for θ because its expectation, $n\theta$, depends on θ.

B. T is sufficient for θ because the law of the sample given $T = t$ is multinomial $(t; 1/n, \ldots, 1/n)$, free of θ.

C. T is sufficient for θ only for large n, when $\overline{X}$ is approximately normal by the CLT.

D. $\overline{X}$ is sufficient for θ but T is not, because only $\overline{X}$ is unbiased for θ.

b) (3 points) The Cramér–Rao lower bound for the variance of unbiased estimators of θ^2 is:

A. θ/n (the bound for θ itself).

B. $4\theta^2/n$ (squared derivative, unit information).

C. $2\theta^3/n$ (derivative not squared).

D. $4\theta^3/n$ (squared derivative over n/θ).

E. $4\theta^3/n^2$ (information counted twice).

c) (2 points, T or F) If an unbiased estimator of $g(\theta)$ attains the Cramér–Rao bound at every θ, then it is the UMVU of $g(\theta)$.

d) (2 points, T or F) If a UMVU estimator of $g(\theta)$ exists, then it attains the Cramér–Rao bound.

Programming (added, not in the book). For $\theta = 2$, $n = 10$: $T(T-1)/n^2 = \overline{X}^2 - \overline{X}/n$ is unbiased for θ^2 and a function of the complete sufficient T, hence the UMVU. Compute its exact variance by summing over $T \sim \text{Poisson}(n\theta)$ $(t = 0, \ldots, 399)$ and compare it with the bound of (b).

Block 2 — Poisson tests and intervals *(B 6.5, 6.11; original)*

Question 6. B 6.5 — UMP test for a Poisson mean and its power function.
Source: Bolfarine, H. and Sandoval, M. C. (2010), *Introdução à Inferência Estatística*, Chapter 6, §6.7 Exercícios, Exercício 6.5, p. 115 [PDF p. 121]; the book gives no answer — reproduced verbatim.

6.5. Sejam $X_1, \ldots, X_n$ uma amostra aleatória da variável aleatória $X \sim Poisson(\theta)$.
i) Encontre o teste UMP para testar $H_0 : \theta = \theta_0$ versus $H_1 : \theta > \theta_0$.
ii) Seja $\alpha = 0,05$, faça o gráfico da função poder para $\theta_0 = 1$ e $n = 25$ (use o Teorema do limite central).

Programming (added, not in the book). For $\theta_0 = 1$, $n = 25$, $\alpha = 0.05$: (1) find the smallest integer c with $P(T > c \mid \theta_0) \le \alpha$, $T = \sum_i X_i$, and the randomization probability γ that makes the size exactly 0.05; (2) print, for $\theta \in \{1, 1.2, 1.5, 2\}$, the CLT power from (ii), the exact power of the test "reject if $T > c$" and that of the randomized test; (3) plot the three power curves on $\theta \in [0.8, 2.2]$.

Solution:

Question 7. B 6.11 — likelihood ratio test for two Poisson means.
Source: Bolfarine, H. and Sandoval, M. C. (2010), *Introdução à Inferência Estatística*, Chapter 6, §6.7 Exercícios, Exercício 6.11, p. 116 [PDF p. 122]; the book gives no answer — reproduced verbatim.

6.11. Sejam $X_1, \ldots, X_n$ uma amostra aleatória da variável aleatória $X \sim Poisson(\theta_1)$ e sejam $Y_1, \ldots, Y_m$ uma amostra aleatória da variável aleatória $Y \sim Poisson(\theta_2)$ sendo as amostras independentes.
i) Encontre o teste da RVG(aproximado) para testar $H_0 : \theta_1 = \theta_2$ versus $H_1 : \theta_1 \neq \theta_2$.
ii) Sendo $n = 5$, $\sum x_i = 3,8$; $m = 8$; $\sum y_i = 4,8$, qual a sua conclusão a um nível de significância de 5%?

Programming (added, not in the book). Write `lr2(sx, sy, n, m)` returning $-2\log\lambda$ from the two totals. Apply it to the printed data. Then estimate the true size of the approximate test (reject when $-2\log\lambda > 3.841$) for $n = 5$, $m = 8$ by simulating 100,000 pairs of samples under H_0 with $\theta_1 = \theta_2 = 0.66$ and with $\theta_1 = \theta_2 = 3$ (`np.random.default_rng(7)`).

Solution:

Question 8. Original math question — three confidence intervals for a Poisson rate.
Source: Original question (not from a book).

The National Civil Defence records the yearly number of dam incidents in a region, modelled as $X_1, \ldots, X_n$ i.i.d. Poisson(θ); $T = \sum_i X_i$, $z = z_{1-\alpha/2}$.

(a) Derive the Wald interval from the CLT and Slutsky's theorem. Show that it degenerates when $T = 0$ and prove that, for fixed n, its coverage tends to 0 as $\theta \to 0$.

(b) Derive the score interval by inverting $|\overline{X} - \theta| \le z\sqrt{\theta/n}$. Prove that both endpoints are non-negative and find the interval when $T = 0$.

(c) Prove that $P(T \le t; \mu) = P(G_{t+1} > \mu)$ for $T \sim \text{Poisson}(\mu)$ and $G_{t+1} \sim \text{Gamma}(t+1, 1)$. Use it to write the exact (Garwood) interval with gamma quantiles and prove that its coverage is at least $1 - \alpha$ for every θ.

(d) Parameter analysis: sign of the shift of the score interval's centre relative to $\overline{X}$; ratio of the score and Wald half-widths and its limit as $n \to \infty$; limits of the three coverages as $\theta \to 0$.

Programming (added, not in the book). Write `intervals(t, n)` returning the three 95% intervals and `coverage(theta, n)` computing their *exact* coverage by summing $P(T = t)$ over the t whose interval contains θ (no simulation). Print the coverages for $n = 10$ and $\theta \in \{0.1, 1, 5\}$, and the three intervals for the coal data 1851–1891 ($n = 41$, $T = 127$).

Solution:

Question 9. Original conceptual question — tests and intervals for counts (10 points).
Source: Original question (not from a book).

a) (3 points) To test $H_0 : \theta = 5$ against $H_1 : \theta \neq 5$ with a Poisson sample (B, Exemplo 6.5.5), one uses $-2\log\lambda = 2n[\overline{x}\log(\overline{x}/5) - (\overline{x} - 5)]$. For large n, under H_0 this statistic is approximately:

A. χ^2_n, one degree of freedom for each observation in the sample.

B. $N(0, 1)$, because $\overline{X}$ is asymptotically normal under H_0.

C. χ^2_1, the number of free parameters in H_1 minus that in H_0.

D. χ^2_2, because the alternative $\theta \neq 5$ has two sides.

b) (3 points) Over $n = 12$ years a dam recorded no incident ($\overline{x} = 0$). The 95% Wald interval for the yearly rate θ is:

A. $[0, 0]$, a degenerate interval of zero length.

B. [0, 3.84/12], from inverting the score statistic.

C. [0, 1.96/12], from the normal quantile over n.

D. $[-0.16, 0.16]$, from a symmetric normal band.

c) (2 points, T or F) For Poisson data, a UMP test of $H_0 : \theta \le \theta_0$ against $H_1 : \theta > \theta_0$ exists because the likelihood ratio is monotone in $T = \sum X_i$.

d) (2 points, T or F) Because a UMP test exists for the one-sided alternative, a UMP test also exists for $H_0 : \theta = \theta_0$ against $H_1 : \theta \neq \theta_0$.

Programming (added, not in the book). Check (a) by simulation. Draw 100,000 samples of size $n = 50$ from Poisson(5), through their totals $T \sim$ Poisson(250), using the generator

`np.random.default_rng(9)`; compute the statistic of (a) for each and report the fraction above $\chi^2_{1,0.95} = 3.841$.

Block 3 — Two Poisson regimes: equal rates and the change point *(G 3.8, 7.3; original)*

Question 10. G 3.8 — Bayes factor for two typists.
Source: Greenberg, E. (2008), *Introduction to Bayesian Econometrics*, 1st ed., Cambridge University Press, Chapter 3, §3.5 Exercises, Exercise 3.8, p. 40 [PDF p. 55]; the book gives no answer — reproduced verbatim. Table 3.2, to which it refers, is reproduced from §3.2.4, p. 38 [PDF p. 53].

3.8 Suppose the number of typographical errors per page has a Poisson distribution $P(\theta)$. Sam types the first m pages of a manuscript and makes e_1 errors in total, and Levi types the last m pages and makes e_2 errors. Let Sam's error rate be θ_1 and Levi's error rate be θ_2. To compare model $M_1 : \theta_1 = \theta_2 = \theta$ with model $M_2 : \theta_1 \neq \theta_2$, produce a table like Table 3.2 for $m = 100$ with 10 total errors and for $m = 200$ with 20 total errors and Sam's error proportions of 0.9, 0.8, 0.7, 0.6, and 0.5. Take $G(1, 1)$ as a prior for θ_1 and θ_2.

Referred to (Greenberg 2008, p. 38): **Table 3.2.** Bayes Factors for Selected Possible Outcomes

Michaela Proportion heads	Lila Proportion heads	$\log_{10}$(Bayes factor) $m = 10$	$m = 100$
0.1	0.9	−2.506	−30.775
0.2	0.8	−1.200	−15.793
0.3	0.7	−0.348	−6.316
0.4	0.6	0.138	−0.975
0.5	0.5	0.297	0.756

Programming (added, not in the book). Compute the table from the closed form. Then check the cell $m = 100$, $(e_1, e_2) = (9, 1)$ by Monte Carlo integration over the prior: with `np.random.default_rng(10)` draw 10^6 values from $G(1, 1)$ for each of the three integrals (M_1; M_2's two factors), average the integrands, and report $\log_{10} \hat{B}_{12}$ with its delta-method standard error.

Solution:

Question 11. G 7.3 — Gibbs sampler for the coal-mining switch point (computer exercise).
Source: Greenberg, E. (2008), *Introduction to Bayesian Econometrics*, 1st ed., Cambridge University Press, Chapter 7, §7.5 Exercises, Exercise 7.3, p. 106 [PDF p. 121]; the book gives no answer — reproduced verbatim. Equations (7.3) and (7.4), to which the exercise refers, are reproduced verbatim from §7.1.1, pp. 93–94 [PDF pp. 108–109].

Equations referred to (Greenberg 2008, pp. 93–94):

$$p(y_i) = \begin{cases} \dfrac{e^{-\theta_1}\theta_1^{y_i}}{y_i!}, & \text{for } i = 1, \ldots, k, \\ \dfrac{e^{-\theta_2}\theta_2^{y_i}}{y_i!}, & \text{for } i = k+1, \ldots, n, \end{cases} \tag{7.3}$$

"where $y_i = 0, 1, \ldots,$ and the switch point k is unknown. The specification is completed by assigning the conditionally conjugate priors,

$$\theta_1 \sim G(\alpha_{10}, \beta_{10}), \quad \theta_2 \sim G(\alpha_{20}, \beta_{20}), \quad \pi(k = j) = 1/n, \quad j = 1, \ldots, n." \tag{7.4}$$

7.3 Construct a Gibbs algorithm to analyze the Poisson model with unknown switch point. Given the specification in Equations (7.3) and (7.4), show that

$$\pi(\theta_1, \theta_2, k \mid y) \propto \theta_1^{\alpha_{10}-1} e^{-\beta_{10}\theta_1} \theta_2^{\alpha_{20}-1} e^{-\beta_{20}\theta_2} \prod_{i=1}^{k} e^{-\theta_1} \theta_1^{y_i} \prod_{i=k+1}^{n} e^{-\theta_2} \theta_2^{y_i},$$

and verify that

$$\theta_1 \mid y, k \sim G\Big(\alpha_{10} + \sum_{1}^{k} y_i,\ \beta_{10} + k\Big),$$

$$\theta_2 \mid k, y \sim G\Big(\alpha_{20} + \sum_{k+1}^{n} y_i,\ \beta_{20} + n - k\Big),$$

$$\pi(k \mid y, \theta_1, \theta_2) = \frac{e^{k(\theta_2-\theta_1)} (\theta_1/\theta_2)^{\sum_1^k y_i}}{\sum_{k=1}^{n} e^{k(\theta_2-\theta_1)} (\theta_1/\theta_2)^{\sum_1^k y_i}}, \qquad k = 1, \ldots, n.$$

Apply your algorithm to the mining disaster data analyzed by Carlin et al. (1992a); the data may be found in the article. Your analysis should generate marginal posterior distributions for θ_1, θ_2, $p(k)$, and the marginal likelihood computed by the method of Chib (1995). For simplicity, set $\alpha_{10} = \alpha_{20} = 0.5$ and $\beta_{10} = \beta_{20} = 1.0$. (Curiously, although these data have been analyzed from the Bayesian viewpoint in several papers, none of them present historical material that might shed light on when changes in parameters took place. For example, changes in government regulation, technology, or the location of mines might be expected to affect the probability of a disaster. A more general version of the model permits multiple switch points if necessary.)

Programming (added, not in the book). Beyond the book's tasks (use the coal series of the Instructions, `np.random.default_rng(2026)`, start $k^{(0)} = 56$, 1,000 burn-in and $G = 10{,}000$ kept draws): (1) compute the *exact* $p(k \mid y)$ by integrating θ_1, θ_2 out analytically and report the total-variation distance between it and the Gibbs frequencies of k; (2) report the numerical standard error of the posterior mean of θ_1 by batch means (50 batches of 200), the i.i.d. standard error $s/\sqrt{G}$, and their squared ratio (inefficiency factor).

Solution:

Question 12. Original math question — likelihood ratio for a change point.
Source: Original question (not from a book).

Counts $y_1, \ldots, y_n$ are independent with $y_i \sim \text{Poisson}(\theta_1)$ for $i \le \tau$ and $\text{Poisson}(\theta_2)$ for $i > \tau$, $1 \le \tau < n$. Let $S = \sum_{i=1}^{n} y_i > 0$ and $S_\tau = \sum_{i \le \tau} y_i$.

(a) For known τ, derive the MLEs and the statistic $\Lambda_\tau = -2 \log \lambda_\tau$ for $H_0 : \theta_1 = \theta_2$. Show that $\Lambda_\tau = 2S\,\mathrm{KL}(p \,\|\, \pi)$, the Kullback–Leibler divergence between Bernoulli(p) and Bernoulli(π) with $p = S_\tau/S$ and $\pi = \tau/n$, and prove that $\Lambda_\tau \ge 0$ with equality iff $p = \pi$.

(b) Prove that under H_0, conditionally on S, $S_\tau \sim \text{Binomial}(S, \tau/n)$ whatever the common rate, and use it to build an exact test for known τ.

(c) For unknown τ, let $M = \max_{1\le\tau<n} \Lambda_\tau$ (the profile-likelihood statistic) and $\hat\tau$ its argmax. Prove that, under H_0, $P(M > c) \ge P(\Lambda_{\tau_0} > c)$ for any fixed τ_0 and c, and conclude what happens to a test that compares M with the χ^2_1 critical value.

(d) Parameter analysis: for fixed S and π, prove the sign of $\partial\Lambda_\tau/\partial p$ and the convexity of Λ_τ in p; find its limits as $p \to 0$ and $p \to 1$; say how Λ_τ scales with S at a fixed p. Interpret.

Programming (added, not in the book). For the coal series ($n = 112$): compute Λ_τ for $\tau = 1, \dots, 111$, $\hat\tau$ and M; the conditional binomial p-value of (b) at $\tau = \hat\tau$ (and why it is not a valid p-value once τ was chosen by the data); and the null distribution of M by a parametric bootstrap (2,000 series of 112 i.i.d. Poisson(S/n) counts, `np.random.default_rng(12)`): its 95% quantile, the rejection rate of the rule "$M > 3.841$" and the bootstrap p-value of the observed M.

Solution:

Question 13. Original conceptual question — switch points and Bayes factors (10 points).
Source: Original question (not from a book).

a) (3 points) In the Gibbs sampler of Question 11, the full conditional distribution of the switch point k is:

A. uniform on $\{1, \dots, n\}$, its prior, because k does not appear in the gamma conditionals.

B. a gamma distribution, because the gamma prior is conjugate to the Poisson likelihood.

C. Binomial$(S, k/n)$, the conditional law of the first-segment count given the total.

D. discrete on $\{1, \dots, n\}$ with weights proportional to $e^{k(\theta_2-\theta_1)}(\theta_1/\theta_2)^{S_k}$.

b) (3 points) The Bayes factor B_{12} of Question 10 depends on the page-level data only through:

A. the individual page counts, through the product of factorials $\prod_i y_{1i}! \prod_i y_{2i}!$.

B. the totals e_1 and e_2 (given m), because the factorials cancel in the ratio.

C. the grand total $e = e_1 + e_2$, because model M_1 pools the two typists' pages.

D. the proportion e_1/e alone, whatever the number of errors behind it.

c) (2 points, T or F) In the switch-point model with independent gamma priors, θ_1 and θ_2 are independent a posteriori *given* k.

d) (2 points, T or F) Hence θ_1 and θ_2 are also independent in their marginal posterior, after k is integrated out.

Programming (added, not in the book). Using the Gibbs draws of Question 11, compute the sample correlation of $(\theta_1^{(g)}, \theta_2^{(g)})$; compute the exact posterior correlation from the mixture over $p(k \mid y)$ of the moments of the two conditional gammas $\theta_j \mid y, k$ given in the statement of Question 11.

Block 4 — Bayesian gamma–Poisson: posterior, predictive, Bayes estimators *(G 2.2, 3.2, 3.5; B 4.9/5.12; original)*

Question 14. G 2.2 — gamma–Poisson conjugacy.
Source: Greenberg, E. (2008), *Introduction to Bayesian Econometrics*, 1st ed., Cambridge University Press, Chapter 2, §2.5 Exercises, Exercise 2.2, p. 19 [PDF p. 34]; the book gives no answer — reproduced verbatim.

2.2 The Poisson distribution has probability mass function

$$p(y_i \mid \theta) = \frac{\theta^{y_i} e^{-\theta}}{y_i!}, \qquad \theta > 0,\ y_i = 0, 1, \ldots,$$

and let $y_1, \ldots, y_n$ be a random sample from this distribution.

(a) Show that the gamma distribution $G(\alpha, \beta)$ is a conjugate prior distribution for the Poisson distribution.

(b) Show that $\bar{y}$ is the MLE for θ.

(c) Write the mean of the posterior distribution as a weighted average of the mean of the prior distribution and the MLE.

(d) What happens to the weight on the prior mean as n becomes large?

Programming (added, not in the book). Use the coal counts 1851–1891 ($n = 41$) and the prior $G(0.5, 1)$ of G 7.3. Evaluate the unnormalized posterior $\theta^{\alpha+s-1} e^{-(\beta+n)\theta}$ on a grid of 200,001 points in $(0, 10]$ (work on the log scale), normalize it with the trapezoidal rule, and compare its mean with the closed form of (c). Print the prior weight w for $n = 1, 10, 41, 112, 1000$.

Solution:

Question 15. G 3.2 — Poisson–gamma predictive distribution.
Source: Greenberg, E. (2008), *Introduction to Bayesian Econometrics*, 1st ed., Cambridge University Press, Chapter 3, §3.5 Exercises, Exercise 3.2, p. 39 [PDF p. 54]; the book gives no answer — reproduced verbatim.

3.2 Compute the predictive distribution for y_{n+1} if the y_i have independent Poisson distributions with parameter θ, given that the first n experiments yielded $y_1, y_2, \ldots, y_n$ events, respectively. Assume the prior distribution $G(\alpha, \beta)$.

Programming (added, not in the book). For the coal data 1851–1891 with prior $G(0.5, 1)$, simulate the predictive by composition (Greenberg §5.2): draw $\theta^{(g)}$ from the posterior and then $y^{(g)} \sim \text{Poisson}(\theta^{(g)})$, $g = 1, \ldots, 10^6$, with `np.random.default_rng(15)`. Compare the simulated frequencies of $y = 0, 1, 2, 3$, mean and variance with your closed form, and check the closed form at $y = 0, \ldots, 3$ against numerical integration of the predictive integral.

Solution:

Question 16. G 3.5 — the posterior median minimizes expected absolute loss.
Source: Greenberg, E. (2008), *Introduction to Bayesian Econometrics*, 1st ed., Cambridge University Press, Chapter 3, §3.5 Exercises, Exercise 3.5, p. 39 [PDF p. 54]; the book gives no answer — reproduced verbatim.

3.5 Show that the median of the posterior distribution minimizes loss under the absolute value loss function.

Programming (added, not in the book). For the coal posterior $G(127.5, 42)$ of Question 15, write the posterior expected loss $\rho(d) = \int |\theta - d|\, \pi(\theta \mid y)\, d\theta$ with `scipy.integrate.quad`, minimize it with `minimize_scalar` on $[2.5, 3.5]$, and compare the minimizer with the posterior median (`gamma.median`). Print ρ at the median, the mean and the mode, and check numerically that $\rho'(d) = 2F(d) - 1$ at $d = 3$.

Solution:

Question 17. B 5.12 (with B 4.9) — Bayes estimator of $P(X = 0)$ and a Bayesian interval.
Source: Bolfarine, H. and Sandoval, M. C. (2010), *Introdução à Inferência Estatística*, Chapter 5, §5.6 Exercícios, Exercício 5.12, p. 89 [PDF p. 95]; it refers to Exercício 4.9, Chapter 4, §4.5 Exercícios, p. 72 [PDF p. 78], reproduced first. The book gives no answer to either — both reproduced verbatim.

Exercício referido: **4.9.** Em uma área de reflorestamento, o número de árvores de determinada espécie, por hectare, com certa doença tem uma distribuição *Poisson*(θ). A distribuição a priori de θ é exponencial com média igual a 1. Encontre o estimador de Bayes de $P_\theta(X = 0)$ com relação à perda quadrática..

5.12. Considere o Exercício 4.9. Obtenha um intervalo de confiança Bayesiano para θ com coeficiente de confiança $\gamma = 0,95$. Como fica seu intervalo se $x = 4$?

Programming (added, not in the book). For $x = 4$: (1) compute the equal-tailed 95% interval with `scipy.stats.gamma` and check it against the χ^2_{10} quantiles; (2) compute the HPD interval numerically (minimize the length of $[F^{-1}(p), F^{-1}(p + 0.95)]$ over p) and verify that the density is equal at both ends; (3) compute the exact frequentist coverage of the equal-tailed rule $x \mapsto I(x)$ at $\theta \in \{0.5, 1, 4, 8\}$ by summing Poisson probabilities.

Solution:

Question 18. Original math question — Bayes estimators and their frequentist risk.
Source: Original question (not from a book).

An insurer prices a policy on the number of claims per client and year, $X_1, \ldots, X_n$ i.i.d. Poisson(θ), with a prior $\theta \sim G(\alpha, \beta)$ (shape, rate) from past portfolios; $S = \sum_i X_i$.

(a) Derive the Bayes estimator under squared loss, d_1, and under the weighted loss $L(\theta, d) = (\theta - d)^2/\theta$, d_2, stating when d_2 exists. Relate d_2 to a feature of the posterior.

(b) Derive the frequentist risk $R_1(\theta) = E_\theta[(d_1 - \theta)^2]$ and prove that $R_1(\theta) < \theta/n$ (the risk of the MLE $\overline{X}$) exactly on an interval of θ that contains the prior mean α/β; give the interval.

(c) Compute the Bayes risks $r = \int R(\theta)\pi(\theta)d\theta$ of d_1 and of $\overline{X}$ and prove $r(d_1) < r(\overline{X})$.

(d) Parameter analysis: sign of the bias of d_1 as a function of θ; the variance ratio $\mathrm{Var}(d_1)/\mathrm{Var}(\overline{X})$; the limits of the interval of (b) as $n \to \infty$ and as $\beta \to 0$ with α fixed. Interpret for the insurer.

Programming (added, not in the book). With $\alpha = 2$, $\beta = 1$, $n = 10$: print the interval of (b), its $n \to \infty$ limit and the two Bayes risks; with `np.random.default_rng(18)` simulate 400,000 values of S at $\theta \in \{1, 2, 5, 6\}$ and compare the simulated mean squared errors of d_1 and $\overline{X}$ with the formulas.

Solution:

Question 19. Original conceptual question — gamma–Poisson Bayes (10 points).
Source: Original question (not from a book).

An analyst at ANS (the health-insurance regulator) models claims per beneficiary as Poisson(θ) with prior $G(\alpha, \beta)$ and observes n beneficiaries with total s.

a) (3 points) In the posterior mean written as a weighted average of the prior mean α/β and the MLE $\bar{y}$, the weight on $\bar{y}$ is:

A. $n/(\beta + n)$, the share of real observations in the total "sample size".

B. $\beta/(\beta + n)$, the share of the prior's pseudo-observations.

C. $n/(\alpha + n)$, with the prior's events counted as observations.

D. $\alpha/(\alpha + \beta)$, which does not depend on the sample.

b) (3 points) The predictive distribution of a new beneficiary's count y_{n+1} is:

A. Poisson with mean $(\alpha + s)/(\beta + n)$, plugging in the posterior mean.

B. gamma $G(\alpha + s, \beta + n)$, the posterior of θ.

C. negative binomial, with variance larger than its mean.

D. negative binomial, with variance smaller than its mean.

c) (2 points, T or F) Under absolute-error loss the Bayes estimator of θ is the posterior median.

d) (2 points, T or F) Since the Bayes estimator under squared loss is the posterior mean, it is unbiased for θ in repeated sampling.

Programming (added, not in the book). Check (d) with $\alpha = 2$, $\beta = 1$, $n = 10$, $\theta = 1$: with `np.random.default_rng(19)` simulate 400,000 totals $S \sim$ Poisson(10) and report the average of $(\alpha + S)/(\beta + n) - \theta$.

Block 5 — Monte Carlo for inference: integration, importance sampling, accept–reject *(G 5.6, 5.7; MS B.15, B.18; original)*

Question 20. G 5.6 — the GHK importance-sampling estimator (5.3).
Source: Greenberg, E. (2008), *Introduction to Bayesian Econometrics*, 1st ed., Cambridge University Press, Chapter 5, §5.8 Exercises, Exercise 5.6, p. 75 [PDF p. 90]; the book gives no answer — reproduced verbatim. The text it refers to (§5.4, pp. 71–72 [PDF pp. 86–87], including Equation (5.3)) and Equation (5.1) (§5.1, p. 65 [PDF p. 80]) are reproduced first.

Referred to (Greenberg 2008, p. 65): "we generate $U \sim U(0,1)$ and set $U = \frac{F(X)-F(c_1)}{F(c_2)-F(c_1)}$, which implies that

$$X = F^{-1}(F(c_1) + U[F(c_2) - F(c_1)]) \tag{5.1}$$

is a drawing from the truncated distribution."

Referred to (Greenberg 2008, pp. 71–72): "As a second example, we approximate $P(a_1 < X_1 \leq b_1, a_2 < X_2 \leq b_2)$, where $(X_1, X_2) \sim N_2(\mu, \Sigma)$, $\mu = (\mu_1, \mu_2)'$, and $\Sigma = [\sigma_{ij}]$. The desired integral is

$$P(a_1 < X_1 \leq b_1, a_2 < X_2 \leq b_2) = \int_{a_1}^{b_1} \int_{a_2}^{b_2} f(x_1, x_2)\, dx_1\, dx_2,$$

where $f(x_1, x_2)$ is the density function of the bivariate normal distribution. The first step is to rewrite the joint density in marginal-conditional form, $f(x_1)f(x_2|x_1)$ using the results (see Section A.9) that $x_1 \sim N(\mu_1, \sigma_{11})$ and that $x_2|x_1 \sim N(\mu_2', \sigma_{22}')$, where

$$\mu_2' = \mu_2 + (\sigma_{12}/\sigma_{11})(x_1 - \mu_1), \qquad \sigma_{22}' = \sigma_{22} - \sigma_{12}^2/\sigma_{11}.$$

Now let

$$\begin{aligned} y_1 &= (x_1 - \mu_1)/\sqrt{\sigma_{11}}, & y_2 &= (x_2 - \mu_2')/\sqrt{\sigma_{22}'}, \\ a_1' &= (a_1 - \mu_1)/\sqrt{\sigma_{11}}, & a_2' &= (a_2 - \mu_2')/\sqrt{\sigma_{22}'}, \\ b_1' &= (b_1 - \mu_1)/\sqrt{\sigma_{11}}, & b_2' &= (b_2 - \mu_2')/\sqrt{\sigma_{22}'}. \end{aligned}$$

With this transformation, y_1 and $y_2|y_1$ have standard normal distributions, enabling us to write

$$\begin{aligned} P(a_1 < x_1 \leq b_1, a_2 < x_2 \leq b_2) &= \int_{a_1'}^{b_1'} \int_{a_2'}^{b_2'} \phi(y_1)\phi(y_2|y_1)\, dy_1\, dy_2 \\ &= \int_{a_1'}^{b_1'} \phi(y_1) \int_{a_2'}^{b_2'} \phi(y_2|y_1)\, dy_2\, dy_1 \\ &= \int_{a_1'}^{b_1'} \phi(y_1)[\Phi(b_2'|y_1) - \Phi(a_2'|y_1)]\, dy_1. \end{aligned}$$

The algorithm is now implemented by choosing the standard normal distribution truncated to (a_1', b_1') as the importance distribution; that is, draw $Y_1^{(g)}$ from the truncated standard normal using (5.1). With this choice for the importance function, we have

$$P(a_1 < X_1 \leq b_1, a_2 < X_2 \leq b_2) \approx \frac{1}{G} \sum [\Phi(b_2'|Y_1^{(g)}) - \Phi(a_2'|Y_1^{(g)})]\,[\Phi(b_1') - \Phi(a_1')]." \tag{5.3}$$

5.6 Verify Equation (5.3).

Programming (added, not in the book). Use the numbers of G 5.7 (next question). (1) Implement the truncated-normal draw (5.1) and compare the mean of 10,000 draws (`np.random.default_rng(21)`) with `scipy.stats.truncnorm`. (2) Compute by quadrature the

exact variance of one term of (5.3) and of one crude indicator $\mathbf{1}\{a_1 < X_1 \le b_1, a_2 < X_2 \le b_2\}$, their ratio, and check the bound you prove in the solution.

Solution:

Question 21. G 5.7 — importance sampling for a bivariate normal rectangle.
Source: Greenberg, E. (2008), *Introduction to Bayesian Econometrics*, 1st ed., Cambridge University Press, Chapter 5, §5.8 Exercises, Exercise 5.7, p. 75 [PDF p. 90]; the book gives no answer — reproduced verbatim. The method is the one of (5.3), reproduced in Question 20.

5.7 Use importance sampling to compute $P(0.2 < x_1 \le 2,\ -1 < x_2 \le 5)$, where $(x_1, x_2) \sim N_2(\mu, \Sigma)$, $\mu = (1, -0.5)'$, and

$$\Sigma = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}.$$

Programming (added, not in the book). With $G = 10{,}000$ and `np.random.default_rng(21)` (the same truncated-normal draws as Question 20), report the estimate (5.3) and its numerical standard error; compare with the exact value (one-dimensional quadrature of the last integral of the text quoted in Question 20, cross-checked with `multivariate_normal.cdf` and inclusion–exclusion) and with crude Monte Carlo using 10,000 bivariate normal draws from the same generator.

Solution:

Question 22. MS B.15 — generating Poisson values by the inverse transform.
Source: Morettin, P. A. and Singer, J. M. (2021), *Estatística e Ciência de Dados* (versão preliminar, abril 2021), Apêndice B (Noções de simulação), §B.8 Exercícios, Exercício 15, p. 478 [PDF p. 488]; the book gives no answer — reproduced verbatim.

15) Gere 5 valores de uma variável aleatória com distribuição de Poisson de parâmetro $\lambda = 2$.

Programming (added, not in the book). Write `rpois_inv(u, lam)` returning the smallest j with $F(j) > u$, built with the recursion $p_{j+1} = p_j \lambda/(j+1)$, and the number of comparisons it made. With `np.random.default_rng(22)`: generate the 5 values; then 100,000 more, and report their mean, variance, the average number of comparisons, and a chi-square goodness-of-fit test on the classes $0, 1, \ldots, 6, \ge 7$.

Solution:

Question 23. MS B.18 — half-normal by accept–reject, then a normal sample.
Source: Morettin, P. A. and Singer, J. M. (2021), *Estatística e Ciência de Dados* (versão preliminar, abril 2021), Apêndice B, §B.8 Exercícios, Exercício 18, p. 478 [PDF p. 488]; the book gives no answer — reproduced verbatim.

18) Suponha que $Z \sim N(0,1)$ e considere $X = |Z|$. Obtenha a densidade $\pi(x)$ de X. Use o método da aceitação-rejeição para obter uma amostra de $\pi(x)$ com $g(x) = \exp(-x)$. A seguir gere uma amostra de Z, levando em conta que $Z = X$ ou $Z = -X$.

Programming (added, not in the book). With `np.random.default_rng(23)`, draw proposals until 10,000 are accepted; report the empirical acceptance rate against $1/c$, the mean of X against $\sqrt{2/\pi}$, the mean and variance of the signed sample Z, and Kolmogorov–Smirnov p-values of X against the half-normal and of Z against $N(0,1)$.

Solution:

Question 24. Original math question — importance sampling for a rare Poisson tail.
Source: Original question (not from a book).

A hospital's ICU receives $X \sim \text{Poisson}(\lambda)$ emergency admissions per night; the manager needs $p = P(X \geq k)$ for a capacity k well above λ. Monte Carlo is used with G draws.

(a) For the crude estimator $\hat{p} = \frac{1}{G}\sum_g \mathbf{1}\{X^{(g)} \geq k\}$, $X^{(g)} \sim \text{Poisson}(\lambda)$, derive its mean, variance and relative standard error, and prove that for fixed G the relative error tends to infinity as $p \to 0$.

(b) Importance sampling with proposal Poisson(μ): derive the weight $w_\mu(x)$, prove unbiasedness, and derive the exact variance of one term in closed form, $\sigma^2(\mu) = e^{\mu - 2\lambda + \lambda^2/\mu}\, P(Y_{\lambda^2/\mu} \geq k) - p^2$ with $Y_\nu \sim \text{Poisson}(\nu)$.

(c) Prove that $\mu = \lambda$ gives back the crude variance, and that the proposal $f_\lambda(x)\mathbf{1}\{x \geq k\}/p$ has zero variance. Why can it not be used?

(d) Parameter analysis: for $\mu > \lambda$, prove that w_μ decreases in x, deduce $\sigma^2(\mu) \leq p\,(w_\mu(k) - p)$, and show that this bound is minimized at $\mu = k$. Interpret.

Programming (added, not in the book). With $\lambda = 2$, $k = 10$, $G = 10^5$ and `np.random.default_rng(24)`: compute p exactly; run the crude estimator and the IS estimator with $\mu = 10$ (in that order, same generator) and print estimates with standard errors; tabulate $\sigma^2(\mu)$ for $\mu \in \{2, 5, 10, 15\}$ from (b), check it by summing $f_\lambda(x)^2/f_\mu(x)$ over $x \geq k$, and find the μ that minimizes $\sigma^2(\mu)$ on a grid.

Solution:

Question 25. Original conceptual question — simulation methods (10 points).
Source: Original question (not from a book).

a) (3 points) In the accept–reject algorithm with $f \le c\,g$, the number of proposals needed to obtain one accepted draw is:

A. geometric with mean $1/c$, the acceptance probability.

B. Poisson with mean c, the envelope constant.

C. geometric with mean c, since each proposal is accepted with probability $1/c$.

D. binomial with parameters c and $1/c$, one trial per unit of envelope.

b) (3 points) The Monte Carlo estimator of $E[h(X)]$ from G i.i.d. draws has standard error:

A. proportional to $1/G$, because averaging divides by G.

B. $\mathrm{sd}(h(X))/\sqrt{G}$, whatever the dimension of X.

C. $\mathrm{sd}(h(X))/\sqrt{G}$ only when X is scalar and h is bounded.

D. growing exponentially with the dimension of X, like quadrature.

c) (2 points, T or F) An importance-sampling estimator is unbiased, yet a proposal with lighter tails than the target can give it infinite variance.

d) (2 points, T or F) Accept–reject cannot be used unless the normalizing constant of the target density is known.

Programming (added, not in the book). Check (a) on the half-normal sampler of Question 23: with `np.random.default_rng(25)` draw 2×10^6 exponential proposals and uniforms, mark the accepted ones, take the first 100,000 acceptances, and compare the mean and variance of the number of proposals per acceptance with c and $c(c-1)$.

Block 6 — The bootstrap *(I 5.2, 5.4; original)*

Question 26. I 5.2 — probability that an observation is in a bootstrap sample.
Source: James, G., Witten, D., Hastie, T., Tibshirani, R. and Taylor, J. (2023), *An Introduction to Statistical Learning with Applications in Python*, Springer, Chapter 5, §5.4 Exercises (Conceptual), Exercise 2, pp. 224–225 [PDF pp. 232–233]; the book gives no answer — reproduced verbatim (the margin index entry "`np.empty()`" is omitted).

2. We will now derive the probability that a given observation is part of a bootstrap sample. Suppose that we obtain a bootstrap sample from a set of n observations.

(a) What is the probability that the first bootstrap observation is *not* the jth observation from the original sample? Justify your answer.

(b) What is the probability that the second bootstrap observation is *not* the jth observation from the original sample?

(c) Argue that the probability that the jth observation is *not* in the bootstrap sample is $(1-1/n)^n$.

(d) When $n = 5$, what is the probability that the jth observation is in the bootstrap sample?

(e) When $n = 100$, what is the probability that the jth observation is in the bootstrap sample?

(f) When $n = 10{,}000$, what is the probability that the jth observation is in the bootstrap sample?

(g) Create a plot that displays, for each integer value of n from 1 to 100,000, the probability that the jth observation is in the bootstrap sample. Comment on what you observe.

(h) We will now investigate numerically the probability that a bootstrap sample of size $n = 100$ contains the jth observation. Here $j = 4$. We first create an array `store` with values that will subsequently be overwritten using the function `np.empty()`. We then repeatedly create bootstrap samples, and each time we record whether or not the fifth observation is contained in the bootstrap sample.

```
rng = np.random.default_rng(10)
store = np.empty(10000)
for i in range(10000):
    store[i] = np.sum(rng.choice(100, replace=True) == 4)
          > 0
np.mean(store)
```

Comment on the results obtained.

Programming (added, not in the book). Let D be the number of *distinct* original observations in a bootstrap sample of size $n = 100$. Derive $E(D)$ and $\mathrm{Var}(D)$ from indicator variables, then check them with 10,000 bootstrap samples drawn with `np.random.default_rng(26)`.

Solution:

Question 27. I 5.4 — bootstrap standard deviation of a prediction.

Source: James, G., Witten, D., Hastie, T., Tibshirani, R. and Taylor, J. (2023), *An Introduction to Statistical Learning with Applications in Python*, Springer, Chapter 5, §5.4 Exercises (Conceptual), Exercise 4, p. 225 [PDF p. 233]; the book gives no answer — reproduced verbatim.

4. Suppose that we use some statistical learning method to make a prediction for the response Y for a particular value of the predictor X. Carefully describe how we might estimate the standard deviation of our prediction.

Programming (added, not in the book). Fit the Poisson regression $\log \mu_t = \beta_0 + \beta_1 x_t$, $x_t = (t - 56.5)/10$, to the coal series ($t = 1, \ldots, 112$ for 1851–1962) with `statsmodels` GLM. Predict μ for 1900 ($t = 50$). Compute the delta-method standard error of $\hat{\mu}(1900)$ and the bootstrap standard deviation from $B = 2{,}000$ pairs-bootstrap refits (`np.random.default_rng(27)`), with the 95% percentile interval. Compare the two using the Pearson dispersion statistic.

Solution:

Question 28. Original math question — nonparametric versus parametric bootstrap for a Poisson mean.

Source: Original question (not from a book).

Let $x_1, \ldots, x_n$ be counts, $\bar{x}$ their mean and $\hat{\sigma}^2 = \frac{1}{n}\sum_i (x_i - \bar{x})^2$.

(a) Nonparametric bootstrap (X_i^* i.i.d. from the empirical distribution): find $E^*(\overline{X}^*)$ and $\mathrm{Var}^*(\overline{X}^*)$ exactly, without simulation.

(b) Parametric bootstrap (X_i^* i.i.d. Poisson($\bar{x}$)): find $\mathrm{Var}^*(\overline{X}^*)$. If the X_i are i.i.d. with mean θ and variance σ^2, prove $E(\hat{\sigma}^2) = \frac{n-1}{n}\sigma^2$ and compare the expectations of the two bootstrap

variances with the true $\mathrm{Var}(\overline{X})$, both when the Poisson model holds and when $\sigma^2 = \phi\theta$ with $\phi \neq 1$.

(c) For $\psi = e^{-\theta}$ and $\hat{\psi} = e^{-\bar{x}}$, find the parametric-bootstrap bias estimate $\hat{b} = E^*(e^{-\overline{X}^*}) - e^{-\bar{x}}$ in closed form and the bias-corrected estimator.

(d) Parameter analysis: prove $\hat{b} > 0$, give its leading term in $1/n$, and say how it moves with $\bar{x}$. Interpret.

Programming (added, not in the book). For the coal counts 1851–1891 ($n = 41$): print $\bar{x}$, $\hat{\sigma}^2$, the two exact bootstrap variances, $\hat{\sigma}^2/\bar{x}$, $\hat{\psi}$, $\hat{b}$, its leading-term approximation and the corrected estimate. Check them with $B = 20{,}000$ resamples of each kind (`np.random.default_rng(28)`).

Solution:

Question 29. Original conceptual question — the bootstrap (10 points).
Source: Original question (not from a book).

a) (3 points) For large n, the fraction of the original observations that appear at least once in a bootstrap sample of size n is about:

A. 0.5, since each observation is drawn or not with equal chance.

B. $e^{-1} \approx 0.368$, the probability that an observation is never drawn.

C. 1, because the bootstrap sample has the same size as the data.

D. $1 - e^{-1} \approx 0.632$, the complement of never being drawn.

b) (3 points) As $B \to \infty$, the nonparametric bootstrap standard error of $\bar{x}$ equals:

A. $\hat{\sigma}/\sqrt{n}$, with $\hat{\sigma}^2$ computed with divisor n.

B. $s/\sqrt{n}$, with s^2 computed with divisor $n - 1$.

C. $\sqrt{\bar{x}/n}$, the Poisson standard error.

D. 0, because every bootstrap sample has mean $\bar{x}$.

c) (2 points, T or F) For a fixed sample, the bootstrap standard error converges to the true standard error of the estimator as the number of bootstrap replications B grows.

d) (2 points, T or F) With overdispersed counts, a parametric Poisson bootstrap understates the standard error of $\bar{x}$.

Programming (added, not in the book). Illustrate (b)–(c) on the coal counts 1851–1891: with `np.random.default_rng(29)` compute the bootstrap standard error of $\bar{x}$ with $B = 100$, 1,000 and 10,000 resamples and compare them with the $B \to \infty$ value $\hat{\sigma}/\sqrt{n}$.

Block 7 — MCMC: invariance, Metropolis–Hastings, accept–reject versus MH *(G 6.8, 7.4, 7.6; original)*

Question 30. G 6.8 — an invariant density for a continuous-state kernel.
Source: Greenberg, E. (2008), *Introduction to Bayesian Econometrics*, 1st ed., Cambridge University Press, Chapter 6, §6.5 Exercises, Exercise 6.8, p. 89 [PDF p. 104]; the book gives no answer — reproduced verbatim.

6.8 (Bhattacharya and Waymire, 1990, p. 216) Show that

$$\pi(y) = 2(1-y), \qquad 0 \le y \le 1$$

is an invariant density for the transition kernel

$$p(x,y) = \begin{cases} e^{1-x}, & \text{if } y < x, \\ e^{1-x} - e^{y-x}, & \text{if } y > x. \end{cases}$$

Programming (added, not in the book). (1) Check by quadrature that $\int_0^1 p(x,y)\,dy = 1$ at $x = 0.3, 0.7$ and that $\int_0^1 \pi(x)p(x,y)\,dx = \pi(y)$ at $y = 0.1, 0.5, 0.9$. (2) Derive the c.d.f. of $p(x,\cdot)$, simulate the chain by inversion (`brentq` on the branch $y > x$) from $x_0 = 0.5$ with `np.random.default_rng(30)`, discard 1,000 draws, keep 20,000, and compare the sample mean and $P(Y < 0.5)$ with their values under π; report the batch-means n.s.e. and the lag-1 autocorrelation.

Solution:

Question 31. G 7.4 — the acceptance probability of Algorithm 7.5.
Source: Greenberg, E. (2008), *Introduction to Bayesian Econometrics*, 1st ed., Cambridge University Press, Chapter 7, §7.5 Exercises, Exercise 7.4, p. 106 [PDF p. 121]; the book gives no answer — reproduced verbatim. Algorithm 7.5 and Equation (7.6), to which it refers, are reproduced from §7.2, pp. 98–99 [PDF pp. 113–114].

Referred to (Greenberg 2008, p. 98):

$$\alpha(x,y) = \begin{cases} \min\left\{\dfrac{f(y)q(y,x)}{f(x)q(x,y)}, 1\right\}, & \text{if } f(x)q(x,y) \neq 0, \\ 0, & \text{otherwise.} \end{cases} \tag{7.6}$$

Referred to (Greenberg 2008, p. 99): "**Algorithm 7.5: MH for Beta(3, 4) with U(0, 1) proposal**

7.1 Set $x^{(0)}$ equal to a number between zero and one.

7.2 At the gth iteration, generate U_1 and U_2 from $U(0,1)$.

7.3 If

$$U_1 \le \alpha(x^{(g-1)}, U_2) = \frac{U_2^2(1-U_2)^3}{(x^{(g-1)})^2(1-x^{(g-1)})^3},$$

set $x^{(g)} = U_2$. Otherwise set $x^{(g)} = x^{(g-1)}$.

7.4 Go to 2 and continue until the desired number of iterations is obtained."

7.4 Derive $\alpha(x,y)$ for Algorithm 7.5.

Programming (added, not in the book). Run Algorithm 7.5 from $x^{(0)} = 0.5$ with `np.random.default_rng(31)`, discard 500 iterations and keep 5,000 (Greenberg reports an acceptance rate of 0.57 and a mean of 0.4296). Report your acceptance rate and mean against the true

mean 3/7, and compute the exact long-run acceptance probability $\iint f(x)\min\{f(y)/f(x),1\}\,dx\,dy$ by two-dimensional quadrature.

Solution:

Question 32. G 7.6 — accept–reject versus Metropolis–Hastings for a beta mean.
Source: Greenberg, E. (2008), *Introduction to Bayesian Econometrics*, 1st ed., Cambridge University Press, Chapter 7, §7.5 Exercises, Exercise 7.6, p. 106 [PDF p. 121]; the book gives no answer — reproduced verbatim. Algorithm 5.2 and the condition on c, to which it refers, are reproduced from §5.3, p. 66 [PDF p. 81].

Referred to (Greenberg 2008, p. 66): "The accept–reject (AR) algorithm can be used to simulate values from a density function $f(\cdot)$ if it is possible to simulate values from a density $g(\cdot)$ and if a number c can be found such that $f(Y) \le cg(Y)$, $c \ge 1$, for all Y in the support of $f(\cdot)$. [...]
Algorithm 5.2: Accept–reject method

5.1 Generate a value y from $g(\cdot)$.

5.2 Draw a value u from $U(0,1)$.

5.3 Return y as a draw from $f(\cdot)$ if $u \le f(y)/cg(y)$. If not, reject it and return to step 1. (The effect of this step is to accept y with probability $f(y)/cg(y)$.)"

7.6 Estimate the mean of a Beta(3.7, 4.8) distribution with (1) an AR algorithm and a Beta(4, 5) proposal density (you will need to determine the value of c needed in Algorithm 5.2); (2) an MH algorithm with a Beta(4, 5) proposal density. After the break-in sample, graph the values of the mean against the iteration number to monitor convergence. Compare your answers to the true value.

Programming (added, not in the book). With `np.random.default_rng(32)`: (1) evaluate f/g near 0 and at 0.5 for the Beta$(4,5)$ proposal; (2) run AR with a proposal for which a finite c exists (justify it) until 10,000 draws are accepted; (3) run the independence MH with Beta$(4,5)$ (1,000 burn-in, 10,000 kept), report the acceptance rate, the running mean at 1,000, 2,500, 5,000 and 10,000 kept iterations, and the batch-means n.s.e.; plot the running mean.

Solution:

Question 33. Original math question — random-walk MH on the log scale for a Poisson rate.
Source: Original question (not from a book).

The coal posterior of Question 14 is $\pi(\theta \mid y) \propto \theta^{a-1}e^{-b\theta}$ with $a = 127.5$, $b = 42$. Pretend it is not recognized as a gamma and sample it by Metropolis–Hastings with the multiplicative random walk $\theta' = \theta e^u$, $u \sim N(0, s^2)$.

(a) Derive the proposal density $q(\theta, \theta')$ and the acceptance probability. Show that the same probability arises from a symmetric random walk on $\varphi = \log\theta$ with the target written in φ.

(b) Prove that the MH kernel (with its rejection mass) leaves π invariant, via detailed balance.

(c) Suppose the factor θ'/θ is forgotten. Find the distribution the chain then samples and the bias in the posterior mean.

(d) Parameter analysis: prove the limits of the acceptance probability as $s \to 0$ and $s \to \infty$ and explain why the numerical standard error is large at both extremes.

Programming (added, not in the book). Implement the sampler (start $\theta = 1$, 2,000 burn-in, 50,000 kept, `np.random.default_rng(33)` for each run) for $s \in \{0.02, 0.2, 1\}$, reporting acceptance rate, posterior mean and batch-means n.s.e. (50 batches) against the exact mean a/b; repeat $s = 0.2$ without the Jacobian factor.

Solution:

Question 34. Original conceptual question — MCMC (10 points).
Source: Original question (not from a book).

a) (3 points) In MH with a symmetric random-walk proposal, $q(x, y) = q(y, x)$, the probability of accepting a move from x to y is:

A. $\min\{f(y)/f(x), 1\}$, the ratio of target densities capped at one.

B. $\min\{f(x)/f(y), 1\}$, the ratio of current to proposed density.

C. $f(y)/\{f(x) + f(y)\}$, the share of the proposed point in the pair.

D. 1, because a symmetric proposal needs no correction.

b) (3 points) The Gibbs sampler is a special case of MH in which:

A. the proposal is a symmetric random walk on each block in turn.

B. every proposal is accepted with probability 1/2.

C. the draws are independent, as in accept–reject.

D. each block is proposed from its full conditional, and the acceptance probability is 1.

c) (2 points, T or F) If a kernel satisfies detailed balance with respect to π, then π is an invariant distribution of the kernel.

d) (2 points, T or F) Detailed balance is necessary for π to be invariant: every MCMC kernel with invariant distribution π is reversible.

Programming (added, not in the book). Check (d) exactly on a toy target: for $p(x, y)$ on $\{0, 1\}^2$ with $p(0, 0) = 0.3$, $p(0, 1) = 0.2$, $p(1, 0) = 0.1$, $p(1, 1) = 0.4$, build the 4×4 transition matrices P_x (update x from $p(x \mid y)$) and P_y (update y from $p(y \mid x)$), the systematic scan $P = P_x P_y$, and report $\max|\pi P - \pi|$ and $\max_{i,j}|\pi_i P_{ij} - \pi_j P_{ji}|$.